(1). Executive Summary:

This project supports a national healthcare consultancy company's need for finding a way to support health decisions in a practical way. Goals: 1. Screen for diabetes without confirmatory reports of clinical lab use; 2. Detect main factors for life expectancy for countries; and 3. Critically ill hospital patients' survival risk estimation. Three healthcare datasets were viewed for this: 1. CDC survey; 2. WHO country-level data; and 3. 5 U.S. hospitals critically ill in-patient data. As such, uniform look for EDA and machine learning approaches were conducted for given datasets, having primary model being Gradient Boosting with established success criteria.

Consequently, the outcomes were: 1. Diabetes model could point to those likely diabetic or not, from general health, blood pressure, body mass index, and age, without clinical lab work. However, the data could not tell if someone is prediabetic easily, an important stage in preventing diabetes with lifestyle changes. SMOTE oversampling technique did not reveal better results for identifying prediabetes; instead, it showed that it is about feature overlap rather than issues with models. 2. Life expectancy was strongly linked to national income and education for longevity in a country, indicative of need for long-term investments in these areas. Gradient Boosting performed at R2=0.96 and RMSE less than 2 years. 3. Critically ill hospital patients can have the level of mortality risk estimated with hospital-level analytical approach. SUPPORT2 had AUC=0.962. All success criteria were met across the three datasets.

The consultancy company can support healthcare clients to screen for diabetes at low cost, guide life expectancy policy on education and economic growth, and help hospitals plan treatments and end-of-life care. The diabetes model could be added to the EHR through an API, with SUPPORT2 for bedside care, that adheres to HIPAA rules. The models should also be tested over time for their performance with newer data points.

(2). Introduction & Problem Definitions:

This project is aimed at chronic disease for individual screening, population-level health drivers, and predictions of mortality at clinical level. Chronic conditions have 70% death rates in the US with most of healthcare cost being used for it, 90% of $4.1 trillion.[1] While many Americans live with diabetes, there are millions more with prediabetes. Prediabetes management with lifestyle change can prevent diabetes.[2] Still, prediabetes generally goes undetected without clinical lab results, particularly those in impoverished communities. Life expectancy varies over 30 years from country to country. The indicator can be summed from immunization, education, nutritional factors, and economic status.[3] Understanding key factors are signals that can support resourcing and proper policy measures. Critical care patients face greater than 30% death rates. SUPPORT2 showed that doctors often overestimate survival, resulting in death that can be unnecessarily lengthy and painful. End of life patients would want comfortable care.[4] Having information early on for risk would support steps needed, with proper communication from the medical team.

To make decisions that are from well informed source, machine learning should be used to support predictive health analytics for chosen datasets for disease risks and project goals. Diabetes survey observes data at individual-level risks and hospital records look at clinical information, so they are complementary. The project tackles the problem: healthcare industry's need to use data science tools to assess chronic disease predictors at the individual, population, and clinical settings by applying a unified approach for three datasets. Diabetes can help with preventing disease, life expectancy for planning care, and finally, SUPPORT2 help medical facilities like hospitals guide in-patient care. This can guide the consultancy firm to help them solve different problems for their clients. As such, considerable look into three data is needed.

Collected by CDC in 2015, diabetes dataset consisted of 253,680 responders with no diabetes, prediabetes, and diabetes as target, along with 21 features that were behavioral and demographic in nature.[5] Looking at these features have value in real world with their data points, given that age and high blood pressure are associated with diabetes risk itself, without a need for bloodwork. Life expectancy data was initially sourced from WHO, comprising of 193 countries during 2000 to 2015 time-period. Factors include immunizations, GDP, education, mortality, along with life expectancy as the target (regression). For SUPPORT2, it has 9,105 medical records of patients that fall under critical condition collected from 5 medical centers in the US from 1989 to 1994 for nine diseases, with hospital mortality as a binary target.[6] A healthcare consultancy firm can benefit from this type of ML application with clients like hospitals, insurance companies, and public health organizations for advising needs. Consulting company can have three things in mind: a model to evaluate diabetes risks for screening purposes for preventive measure, factors that drive life expectancy help develop policies using analytical methods, and in-hospital mortality rate prediction using models for end-of-life care.

Questions addressed for research apply to client's goals. Can diabetes behavioral/ demographic features place individuals in a disease status using predictive analysis, given severe class imbalance (84% non-diabetic)? Since approximately 2% are prediabetic, most models would likely just predict large groups and dismiss this group altogether. To address this, SMOTE with oversampling was done. This observed if smaller class could be better identified using adjusting class-weight for imbalance for diabetes. Which features lead to determining life expectancy that are for public health, economics, and immunization, in addition to observing predictors that can be used for interventional means? SUPPORT2: hospital check-in features identified be looked at for hospital mortality and can they be used to reduce the risk of death.

These benchmarks were set upfront to find out if the problem was solved or not. For Diabetes the targets were accuracy over 0.80, macro-F1 over 0.35, balanced accuracy over 0.45, and log loss under 0.45. Macro-F1 and balanced accuracy gained more weight here because of how imbalanced the classes are. Life Expectancy as a regression task had $R^2 > 0.90$ with RMSE under 2.0 years and MAE under 1.5 years. Accuracy was not used here since this is regression. For SUPPORT2 classification the goal was accuracy over 0.85, macro-F1 over 0.80, balanced accuracy over 0.80, and log loss under 0.20. For Diabetes, it had no missing data so not imputed, with no imputed-related bias there. Life Expectancy filling in missing values also did not seem to cause big bias issues. SUPPORT2 is different though, with some features missing over 50% of the data; therefore, SUPPORT2 should be evaluated with caution since filling in this much missing with mean or mode could potentially change the data meaning itself. To prepare for modeling, data was cleaned and then imputed first, then EDA was done. Baseline KNN was used, then regularized regression with Lasso, Ridge, and Elastic Net, followed by logistic regression, SVM, decision trees, gradient boosting, and random forest, all tuned with cross-validated grid search. The model progression of testing was done with purposefulness in mind. First, I started with simple models to see if there were linear relationships. I then moved to regularized models to see if there were relationships among features. Tree-based models were then used to see if patterns not seen otherwise can be detected. This progression of testing made it possible in a comprehensive way to compare models using step-by-step approach.

(3). Data Overview:

Source and Collection Methods: Diabetes was collected by CDC BRFSS as a phone survey in 2015. Life Expectancy was gathered by WHO from 2000-2015 at country-level. SUPPORT2 is from 5 US medical center medical records. All are publicly available online. The Diabetes data came from individual recall of their health over the phone, so likely they are inaccurate recall and information about their health. Life Expectancy is composed of data from 193 countries that has information on mortality and immunization to economic status information. SUPPORT2 is hospital medical records for critically ill patients for 5 US hospitals from 1989 to 1994. Unlike survey results, hospital records would not have same individual bias input as those in surveys and also have older medical information for in-patient care as well. Table 1 shows a side-by-side view of these datasets.

Dataset Overview: Table 1

	Diabetes Survey	**Life Expectancy**	**SUPPORT2**
Source	CDC BRFSS 2015	WHO GHO	UCI / 5 U.S. hospitals
Collection	Phone survey	Country aggregation	Clinical records
Target	Diabetes_012 (3-class)	Life expectancy (yrs)	hospdead (binary)
Data Types	float64 (binary/ordinal)	int, float, object	int, float, object
Rows	229,781	2,938	9,105
Columns	22	22	45
Missing Values	None	Up to 22%	Up to 62%

Data Description: Diabetes data has 229,781 rows and 22 columns. Life Expectancy has 2,938 rows and 22 columns of mixed types. SUPPORT2 has 9,105 rows and 45 columns. Table 2 has the statistics broken down for the three datasets for relevant columns. Diabetes columns are mostly binary or ordinal, while being managed as float64. Many variables are yes or no, so correlations should be interpreted with caution. Whereas, Life Expectancy has combination of integer, float, and object, so certain features like country and status need encoding to make the data ready for modeling. SUPPORT2 has the largest number of columns, and consequently, it needs substantial amount of preprocessing before modeling is done.

Summary Statistics: Table 2

Dataset	**Columns**	**Mean**	**Std**	**Min**	**Median**	**Mode**	**Max**
Diabetes	BMI	28.38	6.61	12	28	27	98
Diabetes	GenHlth	2.51	1.07	1.0	3.0	2.0	5.0
Diabetes	Age	8.03	3.05	1.0	8.0	9.0	13.0
Life Exp.	Life expect.	69.22	9.52	36.3	72.1	73.0	89.0
Life Exp.	GDP	7,483	14,270	1.68	1,767	1.68	119,173

SUPPORT2	age	62.65	15.59	18.0	64.9	72.0	101.8
SUPPORT2	charges	59,996	102,649	1,169	25,024	4,510	1,435,423

Data Dictionary: There are 21 features for Diabetes data, mostly binary. Life Expectancy has economic, health-related, and immunization indicators, with 22 columns. SUPPORT2 has 45 clinical columns. Table 3 lists columns that are most relevant to do effective modeling. Diabetes features collected without any clinical testing using low cost means of survey: HighBP, HighChol, and BMI health-risk features for this disease, along with demographic information. Life Expectancy is composed of health-related data, economic status, and it also includes factors like schooling, that have social impact. SUPPORT2 is used to understand critical care patients to help them during hospital admission using data collected at hospitals to predict mortality in the hospital setting.

Key Columns: Table 3

Dataset	Feature/Column	Description of Column	Value Type
Diabetes	Diabetes_012 (Target)	0=no diabetes, 1=prediabetes, 2=diabetes	3-class target
Diabetes	HighBP, HighChol, BMI	BP, cholesterol (0/1); body mass index	Float/Binary
Diabetes	GenHlth, Age	Health (1-5); age (1-13 category bins)	Ordinal
Diabetes	Smoker, PhysActivity	Smoking status; physical activity (0/1)	Binary
Life Exp.	Life expectancy	National life expectancy in years	Cont. target
Life Exp.	Mortality, GDP, Schooling	Deaths per 1000; GDP per capita; education yrs	Float
Life Exp.	Income composition	HDI income index (0 to 1)	Float
SUPPORT2	hospdead	In-hospital death (0=survived, 1=died)	Binary target
SUPPORT2	dzgroup, sps	Disease group; physiology score	Cat. / Float
SUPPORT2	charges, surv2m	Hospital charges (USD); 2-month survival prob.	Float
SUPPORT2	avtisst, adlp	Avg. TISS score; activities of daily living	Float

Data Quality Assessment: Diabetes data did not have missing values but did have 9.4% duplicate rows (23,899 rows) that were removed. Another issue is that the target has an extreme imbalance problem: 84% with no diabetes, 14% diabetes, and 2% prediabetic, leading to majority class bias. Life Expectancy missingness is moderate, with Population (22.2%), Hepatitis B (18.8%), and GDP (15.2%), which was handled by mean/mode imputation. It has impossible data points for BMI as low as 1.0, a clear indication of data entry error. SUPPORT2 had extreme missingness that was imputed: adlp (activities of daily living score, 62.0%), urine (urine output, 53.4%), glucose (blood glucose level, 49.4%), and totmcst (total medical costs, 38.2%) with a negative value of −102.72, likely a billing adjustment or recording error. Imputation at such levels may weaken relationships among features.

Looking at outliers, it seems they are mostly trailing right tails, rather than standalone extreme outlier values. Diabetes had high BMI going up to 98, and some having self-reported of all 30-days of poor mental health days. These extreme values could be real patients that fall under high-risk category for disease and should not simply be discarded given outlier value trails. Life Expectancy also had high values for GDP and infant deaths, along with SUPPORT2 having hospital bill up to $1.4 million. These values too could be real, not just something random in the dataset. So, decision was made to keep them. Using Gradient Boosting, these outliers could be modeled along with the rest of the datasets because it uses cutoff points for modeling instead of distance.

Ethical Considerations: All data is de-identified. Diabetes relies on self-report, introducing responder bias. SUPPORT2 includes race, sex, and income features that carry demographic risk bias. Life Expectancy aggregates to country-level, potentially hiding within-country disparities. To reduce the impact of these concerns, medical records could be used to check survey data collected. Additionally, SUPPORT2 should be used to test across demographic groups, while Life Expectancy could use more local data collection beyond country level. For any clinical implementation, HIPAA needs to be adhered to.

(4). Data Cleaning, Preprocessing, EDA: Data Cleaning Step: Data was cleaned for models to produce reliable outcomes. Duplicates, missing data, and data quality were dealt with before modeling. Modeling choices dictated cleaning decisions. This was to have features with scales that are similar, with complete feature information, and without errors for modeling.

Diabetes initially had 253,680 rows. Diabetes has 23,899 rows that are duplicated and removed, about 9.4% of the data, resulting in 229,781 rows at the end. No missing values to impute. It has highly imbalanced target: 82.7% are non-diabetic, 15.3% diabetic, and 2.0% pre-diabetic. Class imbalance would impact models, with different responses for each model and this was handled during modeling. Duplicate removal was necessary to remove data collected that were repeated survey responses; removing them helped with inflated values of majority class from being more inflated.

Life Expectancy has 2,938 rows without duplicates, but with significant missing data, namely 22.2% for Population, 18.8% for Hepatitis B, and 15.2% for GDP. Mean and mode imputation was used to fill in this missing data. Mean imputation was used for numeric columns, and mode imputation for categorical features. Imputation was used here instead of deleting missing rows, so I don't end up with a dataset that would be reduced by over 40 percent if I deleted these missing value rows. Additionally, there were some unlikely BMI. Some reported as low as 1.0 and decided to keep them. It could be data entry issue.

SUPPORT2 didn't have duplicates in 9,105 rows but had extensive missing data: adlp (62%), urine (53.4%), glucose (49.4%), and totmcst (38.2%). String missingness set to NaN then filled with mean/mode imputation. Negative values for medical costs (-102.72) remained as likely billing adjustments. At 62% missingness for adlp, this would make accurate relationship between daily functional status and mortality difficult to evaluate; thus, these heavily imputed feature(s) should be analyzed with extra care.

Outlier Check: Outlier check done with boxplots. Results for Diabetes data had BMI over 60, with MentHlth and PhyHlth maxed out at 30 days. These are trailing right tails instead of just having isolated points for extreme outliers. These could be representative of real high-risk sub-populations. For Life Expectancy, GDP was over $100,000 while median was only $1,767, and infant deaths reached 1,800. SUPPORT2 had charges up to $1.4 million. These were kept as real inputs, where Gradient Boosting splits on thresholds rather than just distance. Keeping plausible domain outliers is a good decision when the modeling can manage them without concern.

Feature Engineering: Feature engineering done for various models to perform better. For numeric variables with distance-based models like KNN, SVM, Lasso, Ridge, and Elastic Net, StandardScaler was used to convert every feature to same scale. Categorical variables for Life Expectancy and SUPPORT2 used Ordinal Encoder, while Diabetes did not since it is already numeric.

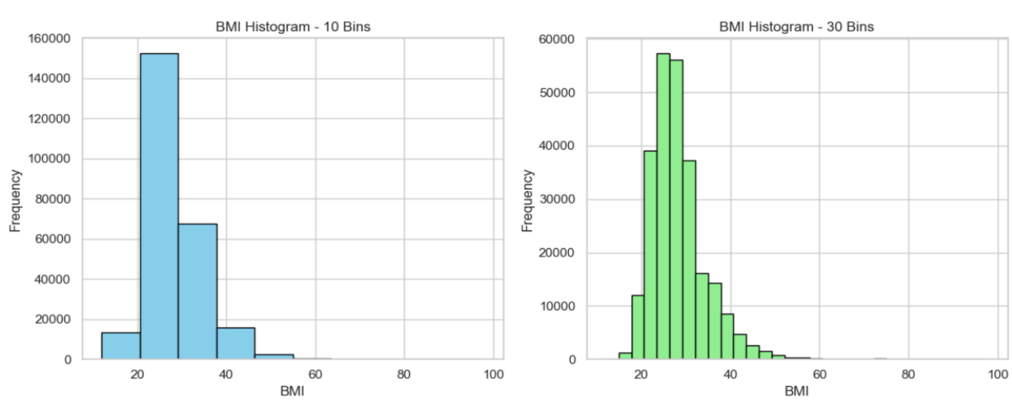
Nonlinear relationships analyzed with degree-2 polynomial and interaction terms. Diabetes had features like BMI and relationship between HighBP and Age generated. Medically this interaction makes sense. As people get older, their blood vessels usually become stiffer, and if they also have high blood pressure, that can put more stress on the body. With time, this can make it harder for the body to regulate sugar level in the blood and can increase insulin resistance. It helps to show that older people with high blood pressure may have a higher diabetes risk. Therefore, these interaction results lead to predictors of this chronic disease or not. VIF analysis showed these new terms created high multicollinearity, so they were only kept inside regularized pipelines. Life Expectancy had naturally high multicollinearity (infant/under-five deaths r=0.997). Diabetes had low multicollinearity without polynomial terms (highest

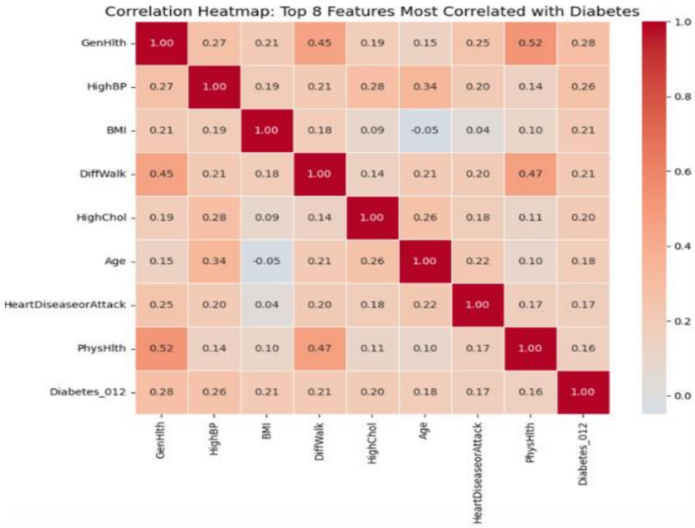
GenHlth at 1.71). No variables were removed because regularized models penalize redundant predictors and tree-based models select valuable features per split.

Dimensionality Reduction: Dimensionality reduction used to see if fewer features would have similar outcome. For Diabetes, forward/backward selection picked 12 of 21 predictors (R2 = 0.17); PCR retained 16 components (87.4% variance, R2 = 0.17); PLSR used 8 components (R2 = 0.17). For Life Expectancy, stepwise selected 12 features (R2 = 0.82), less than the full feature setup. For SUPPORT2, stepwise chose 12 of 45 features (R2 = 0.08). Less features did not enhance performance; fully cleaned data used in modeling. Linear relationships did not work well for this subspace given low R2 values for the targets, which confirms that nonlinear models like Gradient Boosting are the right call here for modeling.

Exploratory Data Analysis (Visualizations): BMI used for histogram: best continuous predictor for diabetes. Fig. 1: BMI distribution looks normal, centered around 28 with long right tails going above 60. Right tail has extreme obesity, high-risk individuals for diabetes screening model. Life Expectancy GDP was highly right skewed (median $1,767 vs. mean $7,483), and SUPPORT2 billing charges reached $1.4 million, justifying the need for scaling. BMI histogram observation indicates that no single threshold can identify diabetes status, which solidifies the rationale for the need for multivariate models, with BMI having some correlation to the target.

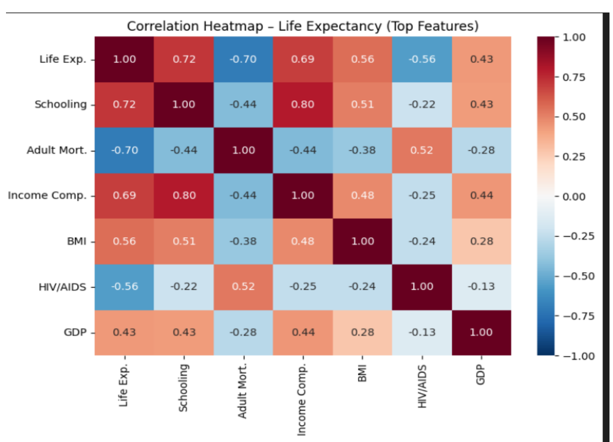
Figure 1: Diabetes BMI Histogram and Correlation Heatmap





Correlation heatmap showed most related features to life expectancy. Fig. 2: Schooling (r = 0.72), Income Composition (r = 0.69), and BMI (r = 0.56) had strong positive relationships with the target. Adult Mortality (r = -0.70) and HIV/AIDS (r = -0.56) with negative correlations. Diabetes feature correlations with the target were weaker, highest being 0.28 for GenHlth. SUPPORT2 strong relationships appeared with surv2m (r = 0.56) and TISS (r = 0.55). There is no single feature that is a strong predictor for diabetes given all correlations are under 0.30; consequently, this supports the use of ensemble modeling for complex interactions detection.

Figure 2: Life Expectancy Correlation Heatmap



Schooling for life expectancy with the strongest correlation. Fig. 3: upper trend with r=0.72, indicating education is connected to health outcomes. Heteroscedasticity: Life expectancy varied below 6 years of schooling, but above 12 years it clustered around 70 to 85 years. Diabetes BMI scatterplot overlapped between target classes. SUPPORT2 surv2m and hospdead were distinct at higher survival levels. Countries with low education show the widest range of life expectancy, educational investment can help with this.

Figure 3: Life Expectancy Scatterplot and Histogram

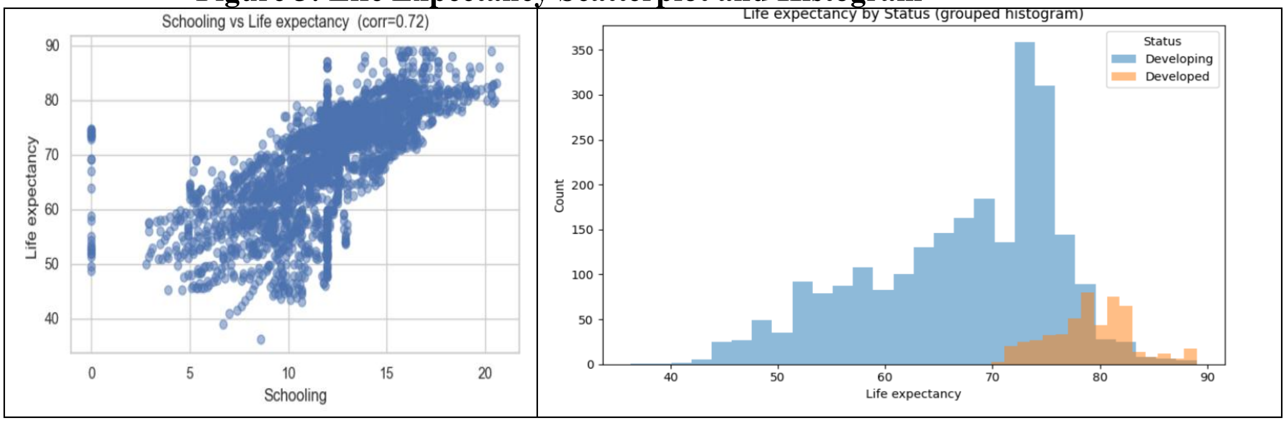
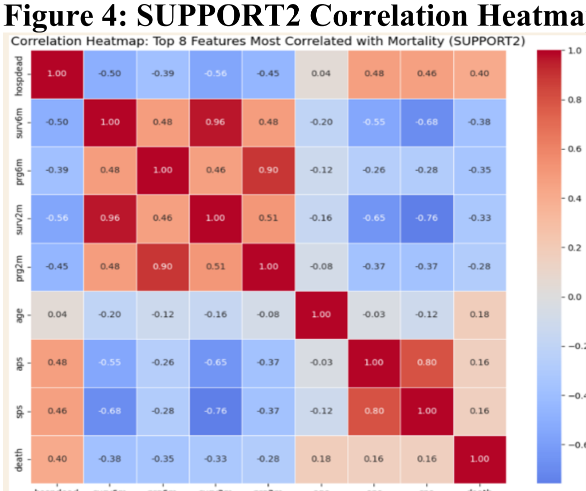


Figure 4: SUPPORT2 Correlation Heatmap



Findings and Summary of EDA: For diabetes, 82.7% majority class; Macro-F1 used for metrics for imbalanced data. Life Expectancy had highly correlated predictors (infants/under-five deaths r=0.997), favoring regularized and tree-based models. SUPPORT2 had missingness up to 62%. This required cautious interpretation. Strongest predictors: GenHlth and HighBP for diabetes, Schooling and Adult Mortality for Life Expectancy, surv2m and avtisst for SUPPORT2. These findings directly shaped modeling decisions. Diabetes with class imbalance, there was justification for balanced class weights and SMOTE implementation, life expectancy has multicollinearity that called for regularized regression to start with, and SUPPORT2 with large amount of imputation required needed careful view of the results.

(5). Modeling/Analysis - Model Choices: A range of models were selected to test out with all three datasets, to learn what works and what does not for these types of data. The models were tested from simple to more complex in a planned fashion. Initially simpler models were used, such as logistic regression for Diabetes and SUPPORT2, and Lasso, Ridge, and Elastic Net for Life Expectancy. This helped to verify simple modeling methods could be used to understand the model performances. Regularized models after that since some features were correlated to each other. Performed tree-based models like decision trees, random forest, and gradient boosting to see if they can be used to observe more complex patterns. Even when a model was not doing well, it was still helpful to compare how a dataset behaved with the modeling.

Regression models implemented were Lasso, Ridge, and Elastic Net, as well as forward/backward selection utilizing PCR and PLSR. Life Expectancy (continuous target) was able to make use of this. To help with feature selection, Lasso was used since L1 penalty can support coefficients to zero. Ridge was also used since L2 penalty reduces coefficient that share relationships rather than just getting rid of them. Finally, with Elastic Net, it could combine both Lasso and Ridge efforts. This applies to Life Expectancy some variables since some variables had high correlation, like for infant deaths and under-five deaths with r = 0.997. For classification models, the following choices were assessed: logistic regression with balanced/unbalanced class weights, support vector machines that use RBF and linear kernels, random forest and decision trees, KNN with variety of distance metrics, and gradient boosting. Both Diabetes, with 3-class target, and SUPPORT2, with binary mortality target, used these items noted here. For clustering, K-Means, DBSCAN, and hierarchical agglomerative clustering were used on all datasets to see whether the resulting clusters matched known clinical groups. The primary model is gradient boosting. This was chosen because it takes advantage of tree-based models, such as the ability to reveal nonlinear relationships, feature relationships, as well as less likely to be influenced by outliers. It can also help with reducing bias to better support predictive power of the model. With sequential build, Gradient Boosting builds tree that corrects errors with each tree from the previous step. Error-correction through sequential means makes it possible for datasets like diabetes to extract minority class incrementally, given its weak signal.

There was a reason why each model was chosen for analysis, for their purposefulness. To see which features were most useful for predicting diabetes, the coefficients from Lasso, Ridge, and Elastic Net were examined, and GenHlth, BMI, and HighBP stood out most as indicators of disease status. For simplicity and easy interpretation, Logistic Regression was used as a baseline model, and it also provided probabilities for clinical risk scoring. With shift from unbalanced to balanced class weights, there was clear trade-off between accuracy and recalling minority-class, resulting in accuracy going from 0.845 to 0.646. SVM was added to see how well a maximum-margin method could handle high-dimensional features. The results showed that SUPPORT2 mortality is linearly separable, with linear kernel of accuracy 0.91, with diabetes having boundaries that are nonlinear for RBF Kernel, which indicate that decision boundaries are fundamentally distinct.

As a baseline model, KNN did not assume the form of decision boundary. It supported testing for patients with similar feature profile could have outcomes that are similar or not. The testing results did reveal that Manhattan outperformed Euclidean distance for Life Expectancy, where R2=0.95 vs. 0.93; L1 distance could deal with skewed data much better. To better gain knowledge of split rules, decision trees were utilized. Whereas random forest was used to support if combination of trees could reduce high level of variance that accompanies a single tree method. Random forest did indeed do better than single trees, which did indicate that bagging led to greater stability. Using SVM, the recall for prediabetics went from 0.00 to 0.33 with balanced class weights. This is clear improvement in performance that showed what adjusting the decision boundary for minority classes can do for modeling performance. With clustering, K-Means, DBSCAN, and HAC, natural patient groups search was done for outside of recognized clinical groups. No such grouping existed, confirming supervised learning use with existing labels.

For Diabetes, SMOTE was used because the prediabetic class was very small, with only about 2% of the data entire dataset. For the training set, roughly 3,700 prediabetic cases were present out of about 184,000 total training rows, so the model could easily disregard this small group. With SMOTE, it creates new minority-class examples with the use of existing minority cases in the feature space. For analysis, a comparison of three cases with Gradient Boosting and balanced class weights were observed: 1. no SMOTE; 2. full SMOTE oversampling, and 3. SMOTE+ENN. The classes were equal for each in size with full SMOTE, while SMOTE+ENN did oversample at the beginning with removal of some noise near other classes for points. Table 4 shows the results.

Table 4: SMOTE Experiment Results (Diabetes)

Approach	**Accuracy**	**Bal. Acc**	**Macro-F1**	**Log Loss**	**Prediab. Recall**
No SMOTE (balanced wt)	0.665	0.506	0.430	0.739	0.213
SMOTE full	0.822	0.405	0.414	0.463	0.000
SMOTE+ENN + balanced	0.778	0.478	0.446	0.590	0.000

Balanced class weights without SMOTE were the only setup helped with any prediabetes occurrence, having a recall at 0.213. Both SMOTE approaches dropped to 0.000 for recall. This happened most likely due to prediabetes overlaps with other classes, both non-diabetes and diabetes, so the new SMOTE points did not support in recognizing separating the groups. This has to do with weak signal in separating the features in the survey dataset, not just class imbalance issue. Consequently, lab for HbA1c or fasting glucose would likely be needed for correctly identifying prediabetes. Although SMOTE did not improve the modeling, it did indicate that minority class prediabetes is difficult to detect.

Training and Validation Strategy: Same training/test split was done for all supervised models, 80% and 20%, respectively. A stratified sampling means was used for classification for class balance. 5-fold stratified cross-validation was then done for training, to assess and tune the models: prevented leakage since test set was isolated for model selection process. Sklearn pipeline was used for scaling, to fit only the training folds without the validation folds. Stratified splitting was used so the small classes, such as for 2% prediabetic group, were in both the train and test sets, to ensure they were in both sets. The pipeline also helped stop data leakage because StandardScaler was fit only on the training data first, then validation folds were also used. This made the model results fairer at the end.

Evaluation Metrics: A look at the problem and quality of data was needed for metric selection. Diabetes and Support2 (classification) used accuracy, macro-F1, balanced accuracy, and log loss. Each metric was used for a valid reason. Accuracy shows overall correct predictions, but it can lead to misunderstanding the results because 82.7% of Diabetes cases are in the majority class. Macro-F1 was useful because it gives each class equal weight, which matters for prediabetes due to imbalance. Balanced accuracy also helped because it checks recall for each class. Log loss was used because probability scores matter more in clinical risk prediction than just a final class label. Life expectancy (regression) used R2, RMSE, and MAE to measure variation in model explanation, how large errors impacted results, and how much the predictions were off in years. R2 shows how much the model explains. RMSE gives more weight to big errors, which matters because being wrong by 10 years for life expectancy is a big problem. MAE is easier to read because it shows the average error in years.

Hyperparameter Tuning: Hyperparameter tuning with grid search and cross-validation was done. Grid search was used instead of random search because there were not too many parameter options to test. It also made the tuning process easier to repeat and compare. For the imbalanced Diabetes and SUPPORT2 classification tasks, macro-F1 was used as the scoring metric inside GridSearchCV so that the search would favor models that performed well across all classes rather than just predicting the majority class. For Life Expectancy regression, RMSE was used as the scoring metric.

For the regularized regression models, I tested alpha on a log scale. The best alpha was very small for Lasso, 0.001, while Ridge worked better with a much larger value, around 345.5. Alpha values were tested on a log scale from 0.001 to 1000 because alpha can have a much different effect depending on how big or small it is. Normal linear grid would waste a lot of tests on bigger values, where the results don't really change much, so it's not the most efficient way testing means. Lasso outperformed others having an alpha of 0.001, so its regularization was minimal to have useful features its purpose. Ridge worked best with a much larger alpha of 345.5 because it shrinks correlated features instead of removing them. This made sense because some Life Expectancy features were highly related to each other. SVM had C, kernel, and gamma method of tuning. Diabetes had best model utilizing RBF kernel with C=1 and gamma=0.1. Whereas SUPPORT2 did best with linear kernel and C=0.1. The C value controls how strict the SVM is with mistakes. A smaller C, like 0.1 for SUPPORT2, gives the model a wider margin, which worked better for the more separable data. For Diabetes, C=1 worked better because the groups were harder to separate. For the RBF kernel, gamma=0.1 controlled how much each training point affected the model. I also tested scale; 0.1 gave better recall for the smaller class.

Step by step tuning gradient boosting (GB) done, started with learning rate (best ones: Diabetes (0.05) and 0.2 for both Life Expectancy and SUPPORT2), then did n_estimators with 100-600, max_depth with 3-5, min_samples_leaf with 10-20, and subsample with 0.9-1.0. I tuned Gradient Boosting step by step because the settings can affect each other. I started with the learning rate because it controls how fast the model learns and how many trees it may need. Diabetes worked best with a lower learning rate of 0.05 because the prediabetes signal was weak, so slower learning made more sense. I kept max_depth at 5 to avoid overfitting and used min_samples_leaf from 10 to 20 so each leaf had enough cases. This was important for diabetes because the prediabetic group was small. For diabetes, step-by-step tuning improved CV log loss from 0.4317 at the starting configuration (learning_rate=0.1, n_estimators=100, max_depth=3) to 0.4283 after tuning all five parameters. The improvement was marginal, which reflects how weak the prediabetes signal is in survey-only features, but documenting each sweep showed that the final configuration was consistent with the best value found at every step. Grid search was also a tuning method for the decision trees and random forest. n_neighbors from 3 to 21 were used to tune KNN, with various distance metrics and weighting approach.

Silhouette scores were used to assess clustering models. K values from 2-10 were tested for K-Means, with diabetes having 0.26 score as the best, Life Expectancy best score of 0.40 and SUPPORT2 best score of 0.17. While DBSCAN and HAC were also tuned, their low silhouette scores suggested that there were no groups other than existing clinical groups. The low silhouette scores across all datasets (all below 0.50) showed that natural clusters are not seen beyond the clinical features that are known. This negative result has meaning and value because it showed that unsupervised approaches could not reveal any hidden subgroups. This confirmed that the supervised learning approach used for the primary analysis is the way to go.

(6). Results:

Model Performance and Success Criteria. Success criteria were met for all. Diabetes didn't have a particular model that won over others. Instead, logistic regression had the leading accuracy with 0.85, KNN with best macro-F1 at 0.39 and SVM with best balanced accuracy of 0.47, and gradient boosting (GB) with the least amount of log loss at 0.43. This is likely due to imbalanced data, with one class having 82.7%. GB did best for probability estimations overall, with other models being best in one area at a time. No single model was the answer across all metrics for diabetes. This difficulty of task indicates that different models are good for different aspects of the imbalanced decision boundary.

Life expectancy had GB as best model: R2=0.96, RMSE=1.91, and MAE=1.23; better performer than regression models, with lower error accompanied by higher accuracy value. GB did better than Ridge, with R2 going from 0.83 to 0.96. This shows that the Life Expectancy data had some non-linear patterns, so tree-based models handled it better than regular regression. Support2 had GB of leading log loss of 0.17, macro-F1 at 0.90, and AUC at 0.962, indicating true death amount above a true survival at 96.2%. SVM had top accuracy at 0.91 with best balanced accuracy at 0.92. Given this, GB did better for metrics that were probability-based. Table 5 Summarizes model results compared with success criteria.

Table 5. Best Model Performance vs. Success Criteria

Dataset	Metric	Target	Achieved	Met?
Diabetes	Accuracy	> 0.80	0.85 (LR)	Yes
Diabetes	Macro-F1	> 0.35	0.39 (KNN)	Yes
Diabetes	Balanced Acc.	> 0.45	0.47 (SVM)	Yes
Diabetes	Log Loss	< 0.45	0.43 (GB)	Yes
Life Exp.	R^2	≥ 0.90	0.96 (GB)	Yes
Life Exp.	RMSE	< 2.0 yrs	1.91 (GB)	Yes
Life Exp.	MAE	< 1.5 yrs	1.23 (GB)	Yes
SUPPORT2	Accuracy	> 0.85	0.91 (SVM/GB)	Yes
SUPPORT2	Macro-F1	> 0.80	0.90 (GB)	Yes
SUPPORT2	Balanced Acc.	> 0.80	0.92 (SVM)	Yes
SUPPORT2	Log Loss	< 0.20	0.17 (GB)	Yes

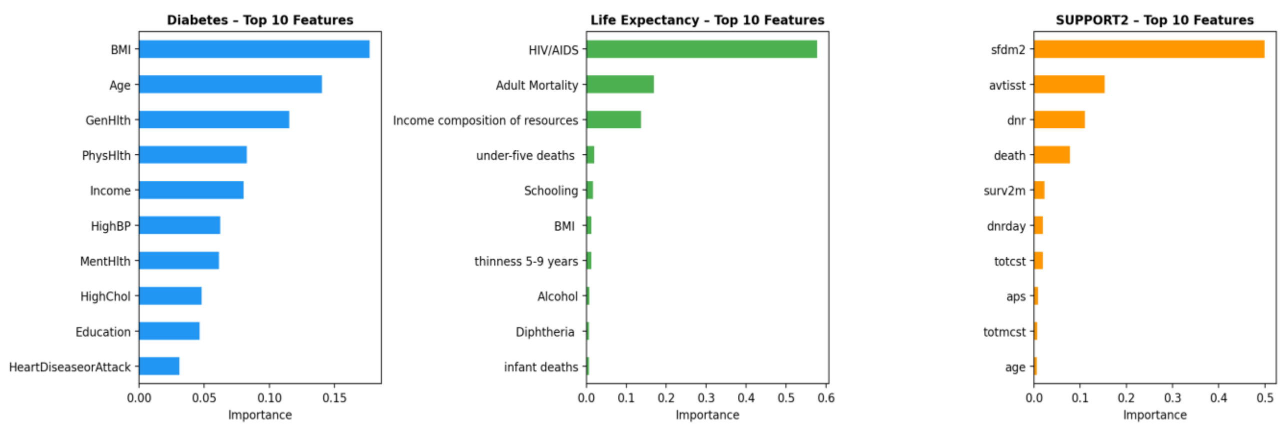
Comparative Analysis of Models: Table 6 compares the models across the three datasets. This made it easier to see which models performed better for each metric. It also helped show that one model was not always best for everything, so the final model choice depended on the dataset and the metric being looked at.

Table 6. Side-by-Side Model Comparison

Model	Diab Acc	Diab F1m	Diab BalA	LE R2	LE RMSE	SUPPORT2 Acc	SUPPORT2 F1m	SUPPORT2 BalA
Logistic Reg	0.85	0.37	0.43	-	-	0.89	0.87	0.88
KNN	0.83	0.39	0.44	0.95	2.20	0.88	0.86	0.87
SVM	0.69	0.42	0.47	0.80	4.13	0.91	0.89	0.92
Decision Tree	0.80	0.34	0.41	0.88	3.31	0.86	0.82	0.84
Random Forest	0.84	0.36	0.42	0.95	2.08	0.89	0.87	0.89
Gradient Boost	0.84	0.38	0.45	0.96	1.91	0.91	0.90	0.91
Lasso/Ridge	-	-	-	0.83	3.84	-	-	-

Feature Importance: Diabetes risk best predictors: GB showed GenHlth, BMI, Age, HighBP, and Income. Life Expectancy important predictors: HIV/AIDS impact, adult mortality, income, and schooling; this indicate that health and social features are key features. SUPPORT2: surv2m, avtisst, sps, and age are most important; health severity is more important than demographic features. Figure 5 presents the top 10 features per dataset. Gradient Boosting used for feature importance; it won over Random Forest in performance, highlighting which predictors mattered most as well. The rankings closely tied to EDA, interestingly. It also showed some additional patterns in this case. For Diabetes, income, a top five feature, while with weak simple correlation. It may be working with other features in a complex way, but correlation does not show for this. For SUPPORT2, surv2m had the strongest predictive ability, to show that the model relies heavily on the 2-month survival estimate, in this case. Without score being available at admission for hospitalization, the model may not perform well as it did.

Figure 5: Gradient Boosting-Feature Importance Top 10 Features



Performance Visualization: Figure 6 AUC: non-diabetic-0.815, Diabetes-0.821, prediabetic-0.691, with SUPPORT2-0.962. The prediabetic AUC was 0.691, less than the non-diabetic and diabetic groups. The model had the most difficult task of separating prediabetes, although it is important for early intervention to prevent diabetes altogether. SUPPORT2 had a much higher AUC of 0.962, indicating better ability of getting mortality risk rank for those admitted. Patients who died had higher risk scores than patients who lived while being admitted. Table 7: Diabetes model had problems with pointing to those with prediabetes, while the SUPPORT2 model correctly showed 1,269 of 1,349 survivors and 389 of 472 those died. Diabetes confusion matrix, the Gradient Boosting didn't identify any prediabetics. All 926 prediabetic patients were predicted as either non-diabetic, 852 cases, or diabetic, 74 cases. This aligns with the SMOTE results as discussed in modeling part of the paper. Prediabetes was hard to pinpoint because the features in the survey unfortunately overlap way to much with the other two groups.

Figure 6. AUC for Diabetes and SUPPORT2

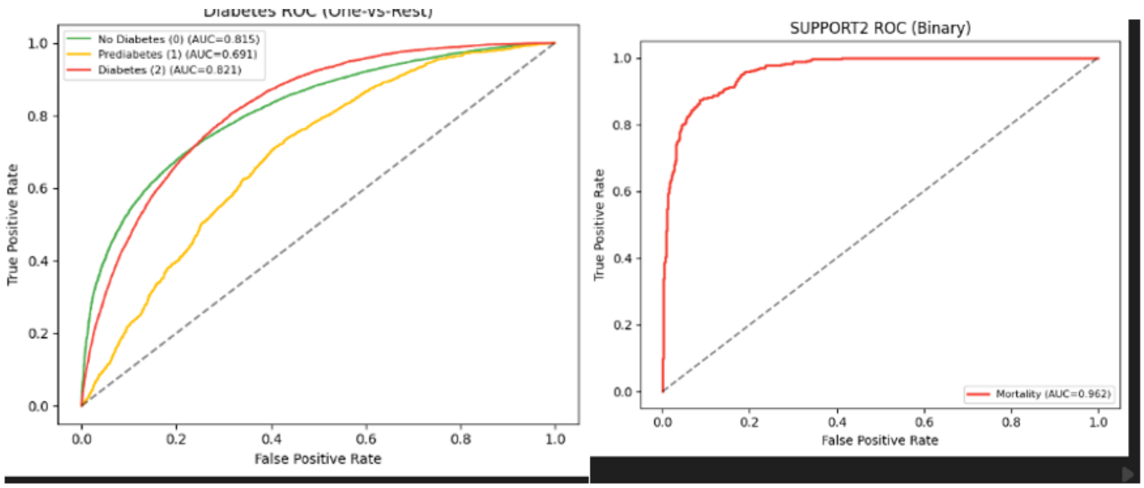


Table 7. Confusion Matrices for Diabetes and SUPPORT2

Diabetes – Gradient Boosting Confusion Matrix

		Predicted 0(No Diabetes)	Predicted 1(Prediabetes)	Predicted 2(Diabetes)	Total
Diabetes	Actual 0 (No Diabetes)	37262	0	750	38012
	Actual 1 (Prediabetes)	852	0	74	926
	Actual 2 (Diabetes)	5807	0	1212	7019

SUPPORT2 – Gradient Boosting Confusion Matrix

		Predicted 0(Survived)	Predicted 1(Died)	Total
SUPPORT2	Actual 0 (Survived)	1269	80	1349
	Actual 1 (Died)	83	389	472

Note: Life Expectancy is a regression task — confusion matrix does not apply. See predicted vs. actual plot (R^2 = 0.967).

Research Questions, Limitations, and Business Implications: Survey data can support diabetes risk understanding based on the results, and education and economic features are closely tied to life expectancy, while variables that are clinical in nature are well placed predictors of mortality in SUPPORT2. Thus, consulting company can utilize modeling to screen diabetes cheaply with this data, use education and economic factors to influence life expectancy outcome, and help hospitals identify high-risk patients early that may lead to death.

There are some limits to these datasets even with best modeling techniques. Prediabetics are not identified because of group overlapping with others; both non-diabetic and diabetic have serious issues because of class imbalance. Life Expectancy: no guarantee best predictors would prolong life at any capacity; it also doesn't have local level data. SUPPORT2 used surv2m/surv6m as predictors. In real world we must assume we have this to do this project. A lot of missing cells lowered the relationships in this case, even with imputing this stands true. Diabetes data is from 2015; models may not work the same now because health/lifestyle changes could be different since then. Life Expectancy is collected at country level, so it can miss local level information. SUPPORT2 is from 1989–1994, so it has outdated hospital care practices. As such, the model should be trained again with data from more current EHR for real setting use.

Diabetes recall of 0.62 indicated that the model performed so that about 62% of true diabetic cases were revealed in the dataset. This can help identify confirmatory lab test candidates for those in higher risk for disease given the modeling capabilities then consulting company clients reduce the unnecessary labs done this way; this saves money by using the screening process as a first step before going to labs. In real use, people flagged by the survey model could be sent for confirmatory lab testing, but additional validation needs to be done for the model before this can be done. For SUPPORT2, the AUC of 0.962 suggests the model ranked risk of death very well. This could help doctors put more focus on higher-risk patients as soon as possible in a hospital setting, without replacing any clinical guidance from medical staff.

(7). Recommendations

Business Actions: The consulting company should use modeling for diabetes data as means to screen for this chronic disease, as a cheaper means than lab work initially as first step. Prediabetes could not be identified here unfortunately with modeling, so those at risk for this disease should be sent for medical testing as a precaution, since this is the only option. The SMOTE results did not really fix the prediabetes issue identification as well. Both SMOTE versions did not capture the existing prediabetic cases. It seems this is occurring because the survey features for prediabetes overlap quite a bit with the non-diabetes and diabetes groups, without clear boundaries. In future work, adding lab values like HbA1c and fasting glucose would probably help separate prediabetes better because that would add clinical values that are normally used for diabetes testing. Life expectancy data should be used to guide policy development to enhance existing education and economic systems to increase lifespan, since these are highly correlated with how long someone lives. SUPPORT2 model should be utilized as a clinical measure to determine mortality risk-level during hospital admission, to direct resource expenditures, treatment plans, and for proper care at end-of-life for admitted patients to provide best care possible for critically ill patients with proper resource allocation where it is needed.

Implementation Considerations: The diabetes data is from 2015 and modeled with more current survey before being implemented in real world setting to capture up-to-date data. SUPPORT2 is from 1989-1994 and should also be retrained on most current EHR before being used in real-world for use. Life expectancy is collected for country-level only, so clients would not be able to develop policy for regional/local need as such, since this is not captured. HIPAA adherence is needed for patient-related medical records for real-world implementation of these models, due to the sensitive nature of these types of healthcare information. Models need to also be updated regularly as new data is added as time goes by to have optimal performance.

For technical deployment, the consulting firm needs to have diabetes screening model to an EHR system using a REST API or public health plugin. Patient survey answers would go into the model, and the model would return a diabetes risk score and predicted class. Those determined to be at high risk can then go for confirmatory lab testing, without the need for testing everyone to save resources. An early screening tool identified through modeling is not meant as final diagnosis for disease status, but rather as an initial look at health status for disease indication. SUPPORT2, the model can be used to help determine mortality risk during when admitted to hospital, particularly useful for first day at ICU. This could be good for planning treatment needs by doctors, when having palliative care needs are warranted, and family discussions about care. It would need HIPAA application as clinical data, that has bank level encryption with access determined by roles. The model needs to be retested periodically on newer data to maintain optimal performance levels. Life Expectancy model can be used as a public health dashboard, with features for schooling, income, GDP, immunization, and mortality to predict life expectancy and show which factors matter most to determine lifespan as accurately as possible. Country-level only is recorded, so newer with local data integration would be needed before using it for policy development/decision that also applies to local levels.

Further Analysis and Data Collection: Diabetes model could benefit with more detailed clinical information, like for prediabetes, since there wasn't enough predictive power for this. It would also enhance the data to support imbalance. SUPPORT2 should have model retraining without survival features, to have a view of how clinical features alone would support the target. Thus, life expectancy should have regional/local data to see how these changes over time could impact the target. These steps would enhance the models' ability to serve usefulness in real-world. For further data collection, I think the firm should focus on three things. First, the diabetes dataset should add lab results like HbA1c and fasting glucose because the survey answers did not separate prediabetes very well. Second, SUPPORT2 should be retrained with newer ICU data from 2020 or later since hospital care has changed since the original dataset. Third, the life expectancy model would be better with state or regional data instead of only country-level data. These changes would make the models more useful in real healthcare and policy settings.

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(1). Executive Summary: (Non-tech paper) This project was conducted to support healthcare consulting firms utilizing chosen datasets to aid its clients in making decisions that are in healthcare arena (hospitals, insurance companies, public health agencies, and policy developers). Three purposes are to address following questions being asked: how to screen for individuals for diabetes without clinical testing for labs, how to comprehend what are the factors that drive life expectancy, and how to have predictive ability to see which in-hospital patients are going to die.

To address these research questions, three datasets were chosen that are available online publicly. Diabetes data is a CDC phone survey of adults, approximately 250,000 respondents.[2] Life expectancy data is from World Health Organization that comprises information collected from 193 countries,[1] while hospital data is from 5 US hospitals for mortality information.[3]

Various prediction tools were used as analysis to test to see which ones do well. The analytical method that did best were those that combined many small decisions. To identify diabetes, responses for general health, BMI, age, and high blood pressure were collected for analysis, while prediabetes could not be identified well. For life expectancy, longevity is correlated to education and lifestyle. For hospital mortality dataset, important items were how sick were the patients when they checked into the hospitals, so the analysis for this data ranked patients by risk for this well, with higher score for death.

Healthcare consulting companies can use analytical approach for: a low-cost measure to screening for diabetes disease or not, a public health policy planning toolkit for life expectancy, and for understanding risk for ICU patients in hospitals. To implement, diabetes and hospital data should be connected to an electronic health records system that is secure and HIPAA compliant. while life expectancy being used through a policy dashboard. Newer data should be used for analysis to support decision making but not replace doctors or clinical testing.

(2). Introduction & Problem Definitions: Business decisions for healthcare organizations are often made without complete information and with insufficient time and financial resources. To help alleviate this issue, this project analyzes three healthcare datasets: diabetes for screening for the disease state, life expectancy for policy planning needs, and hospital data for to understand mortality risk for high-risk patients for mortality. Without lab tests, diabetes cases could go unrecognized. Life expectancy varies from country to country due to health, educational status, economic condition, and disease conditions. Hospital staff need to see warning signs for severely ill patients as soon as they are checked in.

Since this project is targeted for healthcare consulting firm using analytical toolkits for hospitals, insurance companies, and health policy makers, this exercise was done to help its clients make decisions in an informed way. Basically, can the data screen for diabetes risk early-on, what contributes to longevity, and how can you have confirmatory information for in-hospital patients that are at high risk of death. These questions being raised directed this project. The prediction methods approached here need to capture actual diabetes occurrences, country level life expectancy needs to be predicted, and rank hospital mortality risk from low to high risk appropriately. Consequently, the results showed what each analytical tool is capable in terms of predictive abilities: Diabetes data was good for screening diabetes but not prediabetes, so blood draw is needed to check for this. While life expectancy predictive tools were valuable for to understand with key factors like education, income, adult mortality and HIV/AIDS impact. Analytical tools for hospital data worked well; it could separate low and high-risk patients for mortality; this project is good for building tools for understanding the value of these datasets, but it does not replace judgment of healthcare professionals, lab tests, and sound policy choices.

(3). Data Overview: Healthcare consulting firm needs to meet its clients' goals. Three public healthcare datasets were chosen: diabetes for screening disease, public health planning for life expectancy, and hospital mortality risk. The aim wasn't to document each column, rather to convey what matters. Chosen data has mixture of survey results that are yes/no, with categories for disease status and age, along with values for body mass index, life expectancy, GDP, hospital charges, and severity for illness scores.

Diabetes data: CDC phone survey in 2015 for US adults, with about 230,000 respondents and 22 columns. It has diabetes, no diabetes, and prediabetes. Columns that contribute to outcome include general health, body mass index, age, physical activity and income to understand diabetes, without lab work. Life expectancy is from World Health Organization (2000 to 2015) collected from 193 countries, 2,938 rows and 22 columns: each row for one year's worth of data for a country. Life expectancy has columns: for schooling, income, GDP, mortality, immunization rates, and HIV/AIDS impact. Longevity is linked to socio-economic factors, and public health factors. SUPPORT2 data is from records for 9,105 sick hospital patients from 5 US medical centers (1989 and 1994, 45 columns). The goal is to provide whether patients died while hospitalized. Important factors to look at were age, disease group, severity of illness, activity level, hospital bills, and likelihood of survival estimates around admission.

Data has deficiencies. Diabetes data is self-reported; it may be inaccurate, with about 2% prediabetics, so it is hard to detect this group. Life expectancy misses local differences. SUPPORT2 had most missing and older data; it may not capture current hospital practices. Ethically, all datasets were public but de-identified, but real world use still needs to address privacy issues and be HIPAA-compliant systems. The datasets meet project goals, but decisions should be supported with lab tests, doctors' guidance and sound policy decisions.

(4). Data Cleaning and Preparation: Before analysis, datasets had repeated records removed, missing values, and fixable data problem issues were handled for cleaning. Diabetes survey had about 24,000 duplicate responses removed so not to inflate results; it did not have missing values but did have imbalance; most responders didn't have diabetes, with about 2% prediabetic. Further analysis showed that prediabetics weren't easily identified well. Life expectancy data had missing items for population, immunization, and GDP. Gaps were filled out instead of deleting them; removing all incomplete rows could remove important rows. SUPPORT2 had most missing information of three datasets, for activity level, urine- and sugar-related data, so filled them without deleting them to not lose any critical information. Extreme values weren't removed; diabetes data had high body mass index values, life expectancy data had high GDP values, and SUPPORT2 had extremely high hospital bills for patients. These values were kept for more realistic data even if it meant less clean information. Additional prep work was with data for predictive analysis. Text and group-related data like country status or disease group were properly formatted. Very large numeric values were adjusted, so they couldn't overshadow others. Combined some columns like age and high blood pressure to see if that would be helpful for predictions. I used clean data without repeats, missing values, or any data issues since additional steps didn't really help with performance of analysis.

Visual analysis: I used charts like histograms, scatterplots, and relationship tables for columns to understand what would help with analysis. For diabetes, general health, body mass index, age, and high blood pressure were useful for prediction of disease. Life expectancy data had education and adult mortality columns that took center stage. SUPPORT2 showed illness severity was more important than demographics. These items help to direct further analysis and what to recommend.

(5). Analysis: For this part, I did multiple prediction methods, given that the healthcare research questions raised were different for each data. For diabetes, the goal was to group individuals into no diabetes, prediabetes, or diabetes. For hospital data, you wanted to sort people into groups survived or died during a hospital stay. For life expectancy, you wanted to answer how many years people are expected to live in their country.

To accomplish goals of this section, I started with simpler techniques for analysis first. This helped me to understand if there were basic patterns present that would be sufficient to answer the questions raised. I then tried more methods that were flexible to see if more complicated patterns can be detected. Some analytical methods looked at similar people or countries for comparison, while other analytical choices built predictive ability from many small decisions. I also checked to see if data has natural groups to use as a comparison but not as a main analytical approach. Trying multiple methods was fair way to look at choices for analysis.

Diabetes was challenging because prediabetes group was small, so basic way would disregard small group and looked at mostly predicting the largest group. Life expectancy had groups that were related, like education, income, and death rates, so I went with method that would compare multiple patterns at once. For hospital dataset, the approach ranked patients from low to high risk of dying while in the hospital. To test all this, most of the data was used to teach prediction methods, and a separate portion was set aside to test how well they worked on new records. The results are aligned with the goals of the introduction. For diabetes, a check was done to see if the approach used would show actual diabetes cases, not just largest group. Life expectancy, a check was done if predicted years were closely tied to actuals. For hospital data, a check was done to see if high-risk patients ranked higher than low risk, with results next.

(6). Results: The results of the analysis showed that the given data can support healthcare consulting firm's client goals, with varying levels of use and limitations for each approach for this. The best method was using many small decisions, a step-by-step means, prediction approach. This did very well for life expectancy and hospital data, but for diabetes, it was challenging because prediabetes group was small and it overlapped with nondiabetic and diabetic respondents.

For diabetes, the survey is good for first look to screen for diabetes but cannot replace further diagnostic means. The best approach showed that the screening could catch six out of every ten diabetic individuals. This could indicate that community health program and insurance company could look at the survey analysis to see who is a good candidate for blood draw for lab testing could be. This saves money because survey costs less than actual lab work. There is limit to how this could work well given concerns that analytical methods could not predict prediabetics well. This goes beyond the fact that prediabetic group is a small group. A method to aid small group issues was done, but it still was not good enough to solve this concern. The problem is that prediabetic respondents have too many similarities to diabetic and non-diabetic responders with survey answers. This survey can clearly indicate diabetic risk but cannot isolate prediabetics. Lab test for sugar level is needed to confirm the prediabetics.

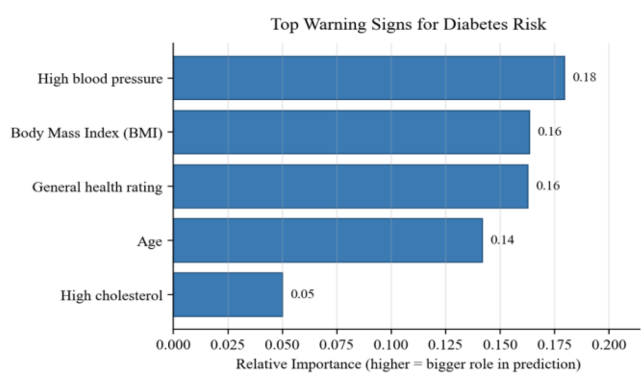
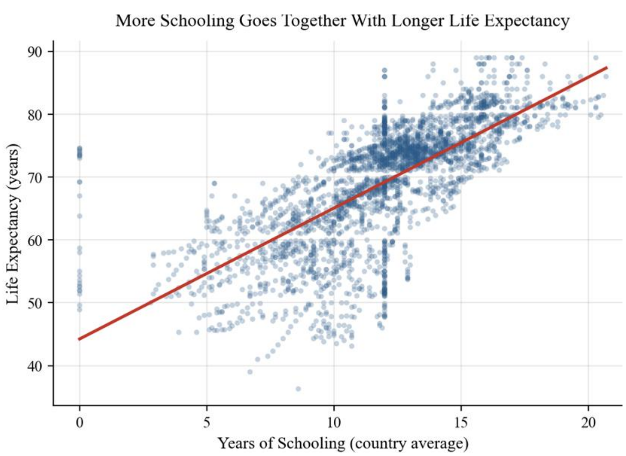


Figure 1: warning indications for diabetes risk

This bar graph of the survey results showed which answer categories were most important to identify risk of diabetes. Bars that were longer mean they played a greater role in this regard. Strongest warning indicators were for general health, body mass index, age, high blood pressure and income. Consequently, Figure 1 is good for diabetes indicator tool, for business application. These factors are the ones that are most important can be collected as surveys, without blood drawing for labs. This toolkit is valuable one for low-cost means for screening diabetes for community health programs, insurance purposes, and for public health outreach. Still, there are limitations of this tool. The consulting firm should use this as screening early on but not as a definitive diagnostic testing method and not for prediabetic identification when marketing this to their clients.

Life expectancy had much better results. The best analytical method predicted a country's life expectancy within about two years on average. This meets the purpose of this dataset for this project. It is a good way to accurate enough means to use for wide range of public health planning needs. The factors that were most important were education, income, adult mortality, and the impact of HIV/AIDS on health at national level. These results seem to align with healthcare conditions and social conditions for life expectancy, which makes sense intuitively.

Figure 2: average years of schooling vs. life expectancy



For Figure 2: Each dot is for a country for one year. Countries that had greater average years of education had longer life expectancy. Countries with more schooling generally have longer life expectancy. This direct relationship can be used to push for education by governments through policy development and investment in this area to help support longer life. This does not show that schooling is the only factor for longer life; however, it does indicate that education has direct relationship to longevity at country level. For the consulting firm, this helps with a policy planning needed by their clients to explore how education, along with income, and health conditions can increase lifespan for countries.

For SUPPORT2, the hospital mortality results were good with analytical methods. The goal was not about just knowing who lived or died while being hospitalized. Instead, it was more about ranking critically ill patients from low- to high-risk while being hospitalized. The analysis did this well. Patients who died basically had higher risk scores than patients who survived while being hospitalized. The group with the highest-risk had a much higher death rate than the one for lowest-risk group according to the results of the best approach for analysis. The business implication is that the consulting firm has three possible services to build from. First, it can offer a low-cost diabetes screening tool that sends high-risk people to confirmatory lab testing. Second, it can offer a public health policy planning tool that shows how education, income, mortality, and disease burden relate to life expectancy. It offers a hospital risk-support tool that helps identify critically ill patients who may need more attention early in admission. The biggest takeaway is that these tools are useful for decision support, but they should not replace lab tests, clinical judgment, or local policy expertise.

Figure 3: hospital death rates vs risk level

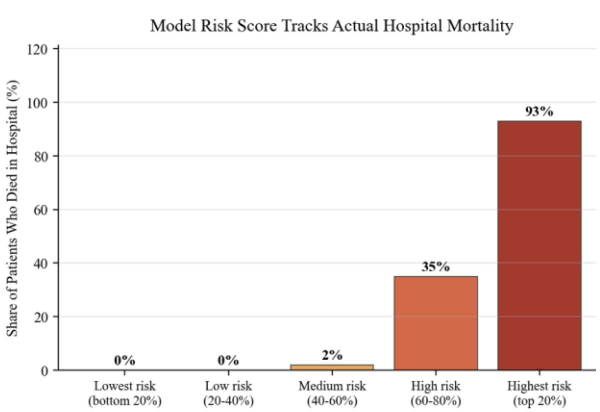


Figure 3: shows that highest-risk patients died more than patients in the lowest-risk group. Risk level matched real-world outcomes well for death in hospital stays. This is good for hospital

clients as a valuable tool. A hospital team can use risk scores to see which patients need closer medical attention. Doctor's judgment is still in order, with care teams a way to notice high-risk patients sooner as well. In critical care, when decisions are made fast, this is a helpful tool.

There are still important limitations. The diabetes data is from 2015 and is based on self-reported survey answers, so the tool should be tested with newer data before real use. The life expectancy data is at the country level, so it cannot explain differences at local level. SUPPORT2 is older hospital data from 1989 to 1994, and some hospital variables had a lot of missing data. Since hospital care has changed, the mortality tool should be retrained on newer electronic health record data before being used in practice in real-world.

The business implication for the consulting firm: 1. It can provide a low-cost diabetes screening tool with high-risk individuals getting to lab test. 2. It can offer public health policy planning guidance for impact of education, income, mortality, and disease burden for lifespan. 3. It can offer a hospital risk-support tool to help determine critically ill patients who may need more attention early when admitted. These tools are helpful to support decision but not to replace lab tests, doctors' recommendations, or any local policy guidance.

(7). Recommendations: The healthcare consulting firm should go ahead with three business goals for this project: diabetes screening toolkit, life expectancy planning toolkit, and a hospital risk-level support tool. These can be provided as means to help support decisions made by its clients.

Diabetes data is good for a low-cost means to first screen for this chronic condition. It can help identify those who are likely in need of follow-up lab testing as a confirmatory process for checking for the disease. This makes it a valuable tool for community health programs, insurance providers to manage cost, and public health outreach to see who needs further testing.

However, it should not be used as a diagnosis and should not be sold as a definitive way to predict prediabetes. The best approach is to do the survey first with individuals, then do the confirmatory lab testing for those who are identified as high risk for this disease based on their answers in the survey.

For life expectancy, the health consulting firm should use the results as for policy planning. Education, income, adult mortality, and the impact of HIV/AIDS were the most important factors that were associated with longer lifespan. Governments, nonprofits, and public health agencies can then use this tool to plan for policy development and updates that align with these key factors to help support longevity. Still, there are limitations of this tool. The findings are based on country-level averages in the dataset, so its use for more, broad implementation for country-level without a look into the local-level data collection that is needed for community-level decision-making considerations.

For hospital mortality, tool used here can be implemented as an early risk-support for hospitals starting at the time of admission for normal part of the facility and for ICU. It can rank admitted patients from low risk to high-risk. Such ranking system would provide early medical review with more attention to higher risk patients, provide appropriate levels of monitoring based on risk, and more timely care planning discussion with family and medical staff. This would support doctors and care teams at the hospital when it comes to patient care and communication with family and caregivers, and consequently, it should not replace clinical judgment.

Before any full implementation of these tools, it would be a good idea to do a small test run for each type of business action as a small pilot first to ensure that any concerns are addressed early on that would minimize any risk of full rollout of these programs. It should also require more current data, for diabetes and hospital tools. Considerations for privacy for individuals and data handling that is secure should be integrated of implementation, along with fair approach for analytical methods. Healthcare consulting firms and their clients should have full and comprehensive understanding of how application of datasets tools is capable of as well as their limitations, as discussed here in this project.

The next step for diabetes is adding lab values such as blood sugar levels. For the hospital tool, another step is retraining it on newer hospital records. For life expectancy, the biggest improvement would be using state-level or regional data instead of only country averages. Overall, the best business move is to offer these as practical tools that improve decisions while being honest about their limits.

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